

Python Assignment 17

1. Create a module named as Arithmetic which contains 4 functions as Add() for addition, Sub() for subtraction, Mult() for multiplication and Div() for division. All functions accept two parameters as number and perform the operation. Write a Python program which calls all the functions from Arithmetic module by accepting the parameters from user.

Answer :

```
import Arithmetic

def main():
    print("Enter Value1")
    Value1 = int(input())
    print("Enter Value2")
    Value2 = int(input())

    Addition = Arithmetic.Add(Value1, Value2)
    print("Addition is :", Addition)

    Subtraction = Arithmetic.Sub(Value1, Value2)
    print("Subtraction is :", Subtraction)

    Multiplication = Arithmetic.Mult(Value1, Value2)
    print("Multiplication is :", Multiplication)

    Division = Arithmetic.Div(Value1, Value2)
    print("Division is : ", Division)

if __name__ == "__main__":
    main()
```

2. Write a program which accepts one number and displays the below pattern.

Input : 5

Output :

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Answer :

```
def main():
    print("Enter the Number :")
    Value = int(input())
    for i in range(Value):
        #print(" * " * Value)
        print(" ".join(["*"] * Value))

if __name__ == "__main__":
    main()
```

3. Write a program which accept one number from user and return its factorial.

Input : 5

Output : 120

Answer :

```
def main():

    fact = 1

    print("Enter the Number : ")
    Value = int(input())

    for i in range(1,Value+1):
        fact = fact * i

    print("Factorial is : ",fact)

if __name__ == "__main__":
    main()
```

4. Write a program which accept one number from user and return addition of its factors.

Input : 12

Output : 16

(1+2+3+4+6)

Answer :

```
def main():
    Add = 0
    print("Enter the Value :")
    Value = int(input())
    for i in range(1,Value,1):
        if Value%i==0:
```

```

        Add+=i
    else:
        i+1
    print("Addition of Factors is ", Add)

if __name__ == "__main__":
    main()

```

5. Write a program which accept one number for user and check whether number is prime or not.

Input : 5

Output : It is Prime Number

Answer:

```

def main():
    Add = 0
    print("Enter the number : ")
    Value = int(input())

    for i in range(1, Value+1, 1):
        if Value%i==0:
            Add += i
        else:
            i+1

    if Add == Value+1:
        print("This is Prime Number")
    else:
        print("This is not a prime Number")

if __name__ == "__main__":
    main()

```

6. Write a program which accept one number and display below pattern.

Input : 5

Output :

```

* * * * *
* * * *
* * *
* *
*

```

Answer :

```
def main():
    print("Enter the number : ")
    Value = int(input())

    for i in range(Value,0,-1):
        print(" * " * i)

if __name__ == "__main__":
    main()
```

7. Write a program which accept one number and display below pattern.

Input : 5

Output : 1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

Answer:

```
def main():
    print("Enter the Value")
    Value = int(input())

    for i in range(Value):
        for j in range(1,Value+1,1):
            print(j, end = " ")
        print()

if __name__ == "__main__":
    main()
```

8. Write a program which accept one number and display below pattern.

Input : 5

Output : 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Answer :

```
def main():
    print("Enter the value")
```

```

Value = int(input())

for i in range(1,Value+1):
    for j in range(1,i+1):
        print(j, end=" ")
    print()

if __name__ == "__main__":
    main()

```

9. Write a program which accept number from user and return number of digits in that number.

Input : 5187934 Output : 7

Answer:

```

def main():
    print("Enter the Value")
    Value = input()
    print(len(Value))

if __name__ == "__main__":
    main()

```

10. Write a program which accept number from user and return addition of digits in that number.

Input : 5187934 Output : 37

Answer :

```

from functools import reduce

def Add(iValue1,iValue2):
    return iValue1+iValue2

def main():
    Values = []
    while True:
        data = input("Enter your numbers : ")

        if data == "":
            break

        Values.append(int(data))

```

```
#if " " in data:
#    Values = list(map(int,data.split()))
#else:
#    Values = list(map(int,data))

Addition = reduce(Add,Values)
print("Addition of all values : ",Addition)

if __name__ == "__main__":
    main()
```